LEARNER GUIDE

The expression behind it

Quadratics. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

You have met a quadratic as a growing sequence and as a constant second difference. Today you meet it as an expression — the symbol with an x^2 in it — and the picture that makes it make sense is area. The area of a rectangle is its width times its height, so $(x + 2)(x + 3)$ is simply the area of a rectangle with sides $x + 2$ and $x + 3$. Split that rectangle where the x ends and the numbers begin, and it falls into four pieces: a square of side x (that is x^2), two thin strips (which make the x terms), and a small rectangle of the two numbers — the constant term (the term with no x in it, just a number on its own). Add the four areas and you have expanded the brackets: $x^2 + (a + b)x + ab$ — add for the middle, multiply for the end. The same picture runs backwards too: given the area (a quadratic like $x^2 + 7x + 12$), you can find the sides — that is factorising, offered here as a stretch when you are ready. There is no calculator today. The question you are exploring is: what does x^2 look like as an area, and can we build it and take it apart?

What you are learning to notice

- That $(x + a)(x + b)$ is the area of a rectangle, and it splits into four pieces: x^2, two x-strips, and the two numbers multiplied.
- That the middle term is the two numbers added, and the constant is the two numbers multiplied.
- That expanding is reading the rectangle forwards (sides $\rightarrow$ area), and factorising is reading it backwards (area $\rightarrow$ sides).
- That you never need to memorise a rule — you can rebuild the expansion from the rectangle every time.

Moving through it, one step at a time

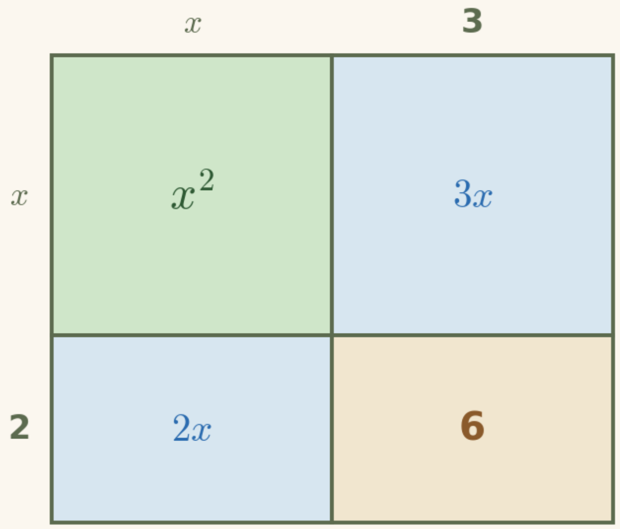
In The rectangle behind the brackets, choose the two numbers and watch the rectangle split into its four areas. In Expand it, fill in $x^2 + \blacksquare x + \blacksquare$ for each product — add for the middle, multiply for the end — using Show the area model when a picture helps. In Spot the slip, catch the two classic errors (multiplying for the middle, or forgetting to multiply the constant). In Factor Finder (a stretch), run it backwards: find the two numbers that multiply to the constant and add to the middle.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — expand with the area model

$(x + 2)(x + 3)$ is the area of this rectangle. Read the four pieces, then join the two x-strips into one middle term.

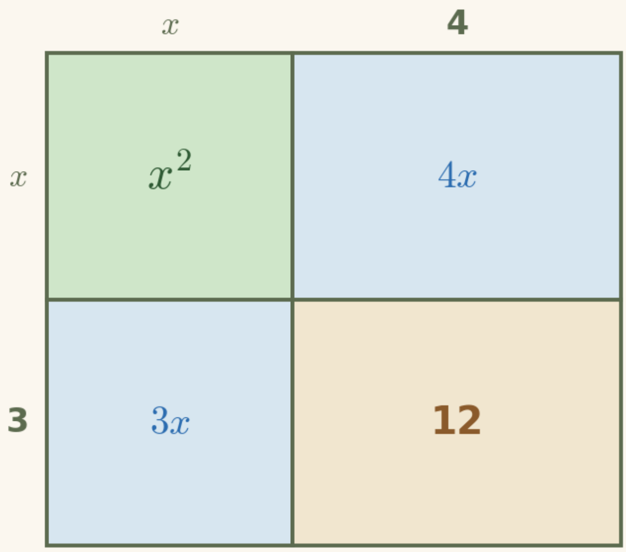


1. the square x^2
2. the two strips join (add) $3x + 2x = 5x$
3. the corner (multiply) $2 \times 3 = 6$

$(x + 2)(x + 3) = x^2 + 5x + 6$

Example 2 — factorise — the model backwards (stretch)

Now you are given the area, $x^2 + 7x + 12$, and you find the sides. You need two numbers that multiply to 12 and add to 7.



1. pairs that multiply to 12 1×12, 2×6, 3×4
2. which pair adds to 7? 3 + 4 = 7
3. so the sides are $(x+3)(x+4)$

$\mathbf{x^2 + 7x + 12 = (x + 3)(x + 4)}$

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the area of a rectangle, multiplying out a single bracket (the distributive law), and collecting like terms), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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